

WASP-39b

R-1438

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_200523 · created 2026-08-01T08:07:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458992.721 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.106 +/- 0.04
Transit depth (Rp/Rs)^2	1.12 +/- 0.86 [%]
Orbital Inclination (inc)	88.16 +/- 1.86
Airmass coefficient 1 (a1)	0.932 +/- 0.013
Airmass coefficient 2 (a2)	0.053 +/- 0.059
Scatter in the residuals of the lightcurve fit is	1.4 %
Best Comparison Star	#3 - [127, 281]
Optimal Aperture	3.73
Optimal Annulus	11.81
Transit Duration (day)	0.103 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.106 ± 0.04 vs 0.1457 (deviation 0.99σ)
Duration fit vs published	0.103 d vs 0.12 d (deviation 0.68σ)
Mid-time O–C	-1.2 min
Depth SNR	2.6
Residual scatter	1.4 %

Light curve

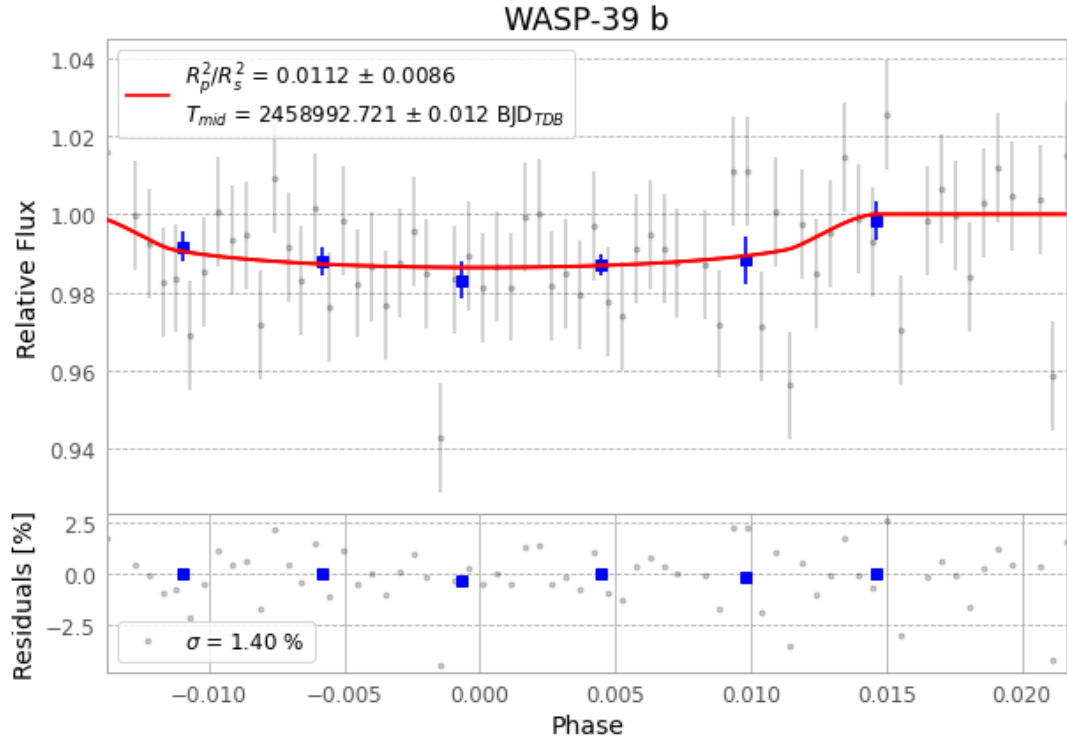


Figure 1 — FinalLightCurve_WASP-39 b_2020-05-22.png (SHA-256 0c3a8742c05c07b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [218, 151], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 909c9b6b24550539...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

69 FITS frames (45 MB), WASP-39200523034815.FITS → WASP-39200523071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-200523.log (49ad4da4ab6e...), FinalLightCurve_WASP-39 b_2020-05-22.png (0c3a8742c05c...), FinalLightCurve_WASP-39 b_2020-05-22.csv (ce0821802517...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 08:07Z.